Seeing chiral superconductivity in QPI

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I. INTRODUCTION

II. BASIC SETUP AND NOTATION

A. Mean field description of superconductors

Superconductors can be modelled by the following (formal) mean-field Hamiltonian:

$$H = \sum_{\boldsymbol{k},ss'} h_{ss'}(\boldsymbol{k}) c^\dagger_{\boldsymbol{k}s} c_{\boldsymbol{k}s'} + \frac{1}{2}\sum_{\boldsymbol{k},ss'} \Big(\Delta_{ss'}(\boldsymbol{k}) c^\dagger_{\boldsymbol{k}s} c^\dagger_{-\boldsymbol{k}s'} + \Delta^*_{ss'}(\boldsymbol{k}) c_{-\boldsymbol{k}s'} c_{\boldsymbol{k}s} \Big) + \mathcal{K} \tag{1}$$

where $h(\boldsymbol{k})$ is the Bloch Hamiltonian of the normal system, we've defined the quantities

$$\Delta_{s_3 s_4}(\boldsymbol{k}) = \sum_{\boldsymbol{k}',s_1 s_2} V_{s_1 s_2 s_3 s_4}(\boldsymbol{k}',\boldsymbol{k}) \, \langle c_{-\boldsymbol{k}' s_2} c_{\boldsymbol{k}' s_1} \rangle \,, \tag{2}$$

$$\Delta^*_{s_1 s_2}(\boldsymbol{k}) = \sum_{\boldsymbol{k}',s_3 s_4} V_{s_1 s_2 s_3 s_4}(\boldsymbol{k}',\boldsymbol{k}) \, \langle c^\dagger_{\boldsymbol{k}' s_3} c^\dagger_{-\boldsymbol{k}' s_4} \rangle \,, \tag{3}$$

$$\mathcal{K} = -\frac{1}{2} \sum_{\boldsymbol{k}\boldsymbol{k}',s_1 s_2 s_3 s_4} V_{s_1 s_2 s_3 s_4}(\boldsymbol{k}',\boldsymbol{k}) \times \langle c^\dagger_{\boldsymbol{k} s_3} c^\dagger_{-\boldsymbol{k} s_4} \rangle \langle c_{-\boldsymbol{k}' s_2} c_{\boldsymbol{k}' s_1} \rangle \,. \tag{4}$$

The BdG Hamiltonian is then:

$$H = \frac{1}{2}\sum_{\boldsymbol{k}} C^\dagger_{\boldsymbol{k}} \hat{H}_{\mathrm{BdG}}(\boldsymbol{k}) C_{\boldsymbol{k}} + \frac{1}{2}\sum_{\boldsymbol{k}} \mathrm{Tr}(\hat{h}(\boldsymbol{k})) + \mathcal{K} \,, \tag{5}$$

$$C_{\boldsymbol{k}} = \begin{pmatrix} c_{\boldsymbol{k}\downarrow} & c_{\boldsymbol{k}\uparrow} & c^\dagger_{-\boldsymbol{k}\downarrow} & c^\dagger_{-\boldsymbol{k}\downarrow} \end{pmatrix}^T \,, \tag{6}$$

$$\hat{H}_{\mathrm{BdG}} = \begin{pmatrix} \hat{h}(\boldsymbol{k}) & \hat{\Delta}(\boldsymbol{k}) \\ \hat{\Delta}^\dagger(\boldsymbol{k}) & -\hat{h}^T(-\boldsymbol{k}) \end{pmatrix} . \tag{7}$$

We ignore the last two terms in from Eqn. (5) here on out, as they not break symmetries (I hope) and correspond to a trivial shift in the BdG energies.

The superconducting mean-field order parameter can be written as a 2×2 matrix in spin-space:

$$\hat{\Delta}(\boldsymbol{k}) = (\Psi(\boldsymbol{k})\hat{\sigma}_0 + \boldsymbol{d}(\boldsymbol{k}) \cdot \hat{\boldsymbol{\sigma}})(i\hat{\sigma}_y) \,, \tag{8}$$

where Ψ is called the spin-antisymmetric (singlet) component and $\boldsymbol{d}$ is the spin-symmetric (triplet) component. When we have no spin-orbit coupling and an SU(2) symmetry due to the spin rotational degree of freedom, the superconducting order parameter is either completely spin singlet or spin triplet (scalar and vector reps of SU(2)). Then, we can enter an invariant spin subspace and treat the BdG Hamiltonian as effectively spinless:

$$C_{\boldsymbol{k}} = \begin{pmatrix} c_{\boldsymbol{k}\uparrow} & c^\dagger_{-\boldsymbol{k}\downarrow} \end{pmatrix}^T \,, \tag{9}$$

$$\hat{H}_{\mathrm{BdG}} = \begin{pmatrix} \varepsilon(\boldsymbol{k}) & \Delta(\boldsymbol{k}) \\ \Delta^*(\boldsymbol{k}) & -\varepsilon(-\boldsymbol{k}) \end{pmatrix} , \tag{10}$$

where we've used $\hat{h}(\boldsymbol{k}) = \varepsilon(\boldsymbol{k})\,\hat{\sigma}_0$ for SU(2) symmetric normal states and that we can assume that either $\hat{\Delta}(\boldsymbol{k}) = \Delta(\boldsymbol{k})(i\hat{\sigma}_y)$ for a spin-singlet superconductor or $\hat{\Delta}(\boldsymbol{k}) = \Delta(\boldsymbol{k})\hat{\sigma}_z(i\hat{\sigma}_y) = \Delta(\boldsymbol{k})\hat{\sigma}_x$ for a spin-triplet state, taking the $\boldsymbol{d}$ vector to point in the $\hat{\boldsymbol{z}}$ direction without loss of generality as all orientations are degenerate.

B. Density of states

C. Quasiparticle Interference

We consider a local impurity in the presence of the superconducting phase defined by the Hamiltonian

$$H = H_0 + V \,, \tag{11}$$

where H_0 is the unperturbed (impurity-free) superconducting Hamiltonian. V is the local impurity term, given as

$$V = \sum_{i\,\sigma\sigma'} [\hat{V}]_{\sigma\sigma'} \delta_{i,\,\boldsymbol{R}=0} c^\dagger_{i\sigma} c_{i\sigma'} \,, \tag{12}$$

where $\hat{V} = V_0\,\hat{I}$ for a scalar impurity and $\hat{V} = V_0\,\hat{\sigma}_z$ for a magnetic impurityThe QPI spectra are calculated using the T-matrix formalism

$$\delta\rho(\omega,\boldsymbol{r}) = -\frac{1}{\pi}\mathrm{Im}[\mathrm{Tr}[\hat{G}_0(\omega,\boldsymbol{r})\,\hat{T}(\omega)\,\hat{G}_0(\omega,-\boldsymbol{r})]] \,, \tag{13}$$

where the T-matrix can be shown to take the form:

$$\hat{T}(\omega) = \frac{\hat{V}}{1 - \hat{G}_0(\omega,\boldsymbol{r}=0)\hat{V}} \,. \tag{14}$$

While this definition of the T-matrix is a formal definition, in practice it can be calculated as the non-trivial solution to the matrix equation

$$(1 - \hat{G}_0(\omega,\boldsymbol{r}=0)\hat{V})\,\hat{T}(\omega) = \hat{V} \,. \tag{15}$$

Note that the Born approximation $\hat{T} = \hat{V}$ corresponds to a first-order approximation of the binomial expansion of the T-matrix. Subgap states (that exist in the superconducting gap) are associated with the poles of the T-matrix (at zero temperature)The condition for the existence of these bound states is

$$\det[1 - \hat{G}_0(\omega,\boldsymbol{r}=0)\,\hat{V}] = 0 \,. \tag{16}$$

Assuming a given unperturbed Green's function (Hamiltonian) and ω, this fixes the value of V_0 in Eqn. (12). Note that in principle this could introduce large V_0 values, so accuracy could potentially be improved by including bands higher in energy now accessible to the impurity states.

III. WHY WE CAN'T SEE CHIRALITY IN (REGULAR) QPI

A. Symmetry of the chiral order parameter

The chiral basis for the superconducting order parameter can be viewed as coming from the one dimensional irreps $E_2^{\pm}$ of the group C_6. The two dimensional irrep E_2 is induced from these irreps by the inclusion of the mirror, which swaps the irreps $E_2^{\pm} \to E_2^{\mp}$. Rotations continue to act diagonally on the chiral functions, adding a phase. Only the C_2 rotation is symmetric without a phase.

The superconducting class is the normal subgroup of $G \times (U(1) \rtimes \{1, R\})$ (R is time reversal operation, $U(1)$ is usual gauge symmetry) under which the order parameter remains invariant. The general symmetry operations of this group can written in the form $\varphi(g)g$, where the dressing $\varphi(g)$ is an element of $U(1) \rtimes \{1, R\} \cong O(2)$. The dressing is a homomorphism $G \to O(2)$, where the kernel of homomorphism is the normal subgroup $H \subseteq G$ containing all elements where the irrep acts as the identity $\chi(E) = \chi(h)$. Note th

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In the chiral basis $f_{\pm}(\mathbf{k}) = (k_x \pm ik_y)^2$, write $\Delta = \eta_+ f_+ + \eta_- f_-$. Under the rotation subgroup C_6, f_+ and f_- span the complex-conjugate one-dimensional irreps E_2^+ and E_2^-: a rotation by θ acts diagonally, $\eta_{\pm} \to e^{\mp 2i\theta}\eta_{\pm}$. The mirrors exchange the chiral components, $\sigma_\phi : (\eta_+, \eta_-) \to (e^{-4i\phi}\eta_-, e^{4i\phi}\eta_+)$, where ϕ is the angle of the mirror line. Since conjugation by a mirror maps E_2^+ to the inequivalent irrep E_2^-, the induced representation $E_2 = \mathrm{Ind}_{C_6}^{C_{6v}} E_2^+$ is irreducible. Its kernel is

$$\ker E_2 = \{h \in C_{6v} : \chi(h) = \chi(E)\} = \{E, C_2\}, \qquad (17)$$

so C_2 acts trivially on every E_2 order parameter.

The full symmetry group of the normal state is $\mathcal{G} = C_{6v} \times O(2)$, where $O(2) \cong U(1) \rtimes \{1, R\}$ contains the gauge transformations $e^{i\alpha}$ and the time-reversal operation R, which acts as $(\eta_+, \eta_-) \to (\eta_-^*, \eta_+^*)$. The superconducting class of a state η is its isotropy subgroup

$$K_\eta = \{(g, u) \in \mathcal{G} : u\, g\, \eta = \eta\}. \qquad (18)$$

States related by $\mathcal{G}$ have conjugate isotropy subgroups, so a class is defined up to conjugacy; K_η is in general not normal in $\mathcal{G}$.

Let $G' \subseteq C_{6v}$ be the set of spatial operations that occur in K_η. Since $U(1)$ acts freely on $\eta \neq 0$, K_η contains no nontrivial pure phase. Two cases arise.

- If K_η contains no element of the form $(E, Re^{i\alpha})$, time reversal is broken and each $g \in G'$ occurs with a unique dressing, $K_\eta = \{(g, \varphi(g)) : g \in G'\}$, with $\varphi : G' \to O(2)$ a homomorphism. The class is written $G'(H)$.
- If K_η contains $(E, Re^{i\alpha})$, a global phase rotation sets $\alpha = 0$. Each g then occurs with exactly one pure-phase dressing $\varphi(g) \in U(1)$. Conjugation by R maps $(g, \varphi(g))$ to $(g, \varphi(g)^{-1})$, so uniqueness forces $\varphi(g) = \pm 1$, and $K_\eta = \{(g, \varphi(g))\} \times \{1, R\}$. The class is written $G'(H) \times R$.

In both cases $H = \ker \varphi$ is the subgroup of operations that leave η invariant without dressing. It is normal in G', and it contains $\ker E_2$, but it can be strictly larger. The quotient $G'/H \cong \varphi(G')$ is a finite subgroup of $O(2)$, and therefore cyclic or dihedral.

B. Irreps of chiral bilinears

IV. WHAT'S NEEDED TO SEE QPI

V. RESULTS

All calculations use the nearest-neighbour triangular lattice with a single impurity in the Born approximation (except in Sec. V D), broadening $\Gamma = 10^{-3}$ and impurity strength $V_0 = 0.1$. The gap scale is $|\Delta| = 0.1$ (in units of the hopping), on a 240×240 $\boldsymbol{k}$-mesh with a 120×120 $\boldsymbol{q}$-mesh. The basis is the spinful Nambu basis $(c_{\boldsymbol{k}\uparrow}, c_{\boldsymbol{k}\downarrow}, c^\dagger_{-\boldsymbol{k}\uparrow}, c^\dagger_{-\boldsymbol{k}\downarrow})$ throughout, and we resolve the LDOS by spin, $\delta\rho_\mu$ with $\mu \in \{0, x, y, z\}$. Each panel is normalised to its own maximum, which is given below it. Single-state figures show only the non-zero channels, and in the combined no-SOC figures a vanishing panel is marked ≈ 0. The impurity potential is cut off in momentum by a Gaussian e^{-q^2/q_0^2}, with q_0 quoted in units of $|\boldsymbol{b}_1|$. Every $\delta\rho(\boldsymbol{q})$ and $\delta\rho(\boldsymbol{r})$ is projected onto the irreps of C_{6v}, $f_\nu = (d_\nu/|G|)\sum_g \chi_\nu(g)\, f(g^{-1}\boldsymbol{q})$, and we quote the weight $\sum |f_\nu|^2 / \sum |f|^2$ of each irrep. We compare four pairing states: s, extended s, $d_{x^2-y^2}$ (nematic, one component of E_2) and $d + id$ (chiral E_2). For each state we show $\omega = 0$ and a frequency just below the gap edge, where the signal is largest. The $d_{x^2-y^2}$ state is nodal and has no gap edge, so for it we use $\omega = 0.15$ instead, about half its maximum gap on the Fermi surface. Table I summarises which spin channels are finite in each case, and their irreps.

A. No spin-orbit coupling

a. Scalar (A_1) impurity. For a scalar impurity, $\hat{T} = V_0\tau_z$. Without SOC all spin channels $\delta\rho_{x,y,z}$ vanish identically, as they must, so only $\delta\rho_0 \equiv \delta\rho$ is shown. All four states are shown in Fig. 1. For s, extended s and

TABLE I. Irreps of the spin-resolved QPI $\delta\rho_\mu(\boldsymbol{q})$ for each system, impurity and pairing. 0 means the channel vanishes. Each pairing is labelled by its own C_{6v} irrep, which for FRG includes spin-symmetric pairings allowed under Rashba SOC. Each entry combines $\omega = 0$ and ω_2 (just below the gap edge, or 0.15 for the nodal $d_{x^2-y^2}$ state). All A_1-irrep impurities hold both in the Born approximation and with the full T-matrix. The † notes where the QPI is instead zero in the Born approximation at $\omega = 0$. The A_2 impurity case was only calculated in the Born approximation. "Im" marks a channel that is purely imaginary in $\boldsymbol{q}$-space; all others are real. Irreps with weight below 10^{-3} are omitted. The last column gives the figures in which that case's QPI is plotted.

System	Impurity	Pairing	ω_2	$\delta\rho_0$	$\delta\rho_x$	$\delta\rho_y$	$\delta\rho_z$	Fig.
no SOC	A_1	A_1: s	0.0980	A_1	0	0	0	1(a)
		A_1: ext. s	0.0098	A_1	0	0	0	1(b)
		E_2: $d+id$	0.2142	A_1	0	0	0	1(c)
		E_2: $d_{x^2-y^2}$	0.1500	A_1 + E_2	0	0	0	1(d)
	A_2	A_1: s	0.0980	0	0	0	0	–
		A_1: ext. s	0.0098	0	0	0	0	–
		E_2: $d+id$	0.2142	$A_1{}^\dagger$	0	0	0	2
		E_2: $d_{x^2-y^2}$	0.1500	0	0	0	0	–
Rashba	A_1	A_1: s	0.0980	A_1	0	0	0	5(a)
		A_1: ext. s	0.0187	A_1	0	0	0	5(b)
		E_2: $d+id$	0.2095	A_1	Im: $E_1{}^\dagger$	Im: $E_1{}^\dagger$	$A_1{}^\dagger$	4
		E_2: $d_{x^2-y^2}$	0.1500	A_1 + E_2	0	0	0	5(c)
Rashba–Hubbard	A_1	A_1: U=0, V=0.9	0.0761	A_1	0	0	0	7
(FRG)		E_2: U=8, V=0, χ=+1	0.0846	A_1	Im: E_1	Im: E_1	A_1	6

$d+id$, $\delta\rho$ is purely A_1 at every frequency (weight 1 to numerical precision). So the chiral state is indistinguishable by symmetry from a fully symmetric gap. The nematic $d_{x^2-y^2}$ state, by contrast, leaves a clear E_2 component. Its weight is 41% at $\omega = 0$ and 65% at $\omega = 0.15$ (Fig. 1(d)). This state is nodal. The zero lines of $\Psi(\boldsymbol{k})$ cross the Fermi surface at four points, near 49°, 131°, 229° and 311°, and the maximum gap on the Fermi surface is 0.30. The smallest $|E|$ on the 240 × 240 mesh (0.0107) only measures how close the mesh comes to a node, and is not a gap. QPI from a scalar impurity therefore sees the rotation-symmetry breaking of a nematic state, but not the time-reversal breaking of a chiral one. This is the statement of Sec. III: the $|\eta_+|^2$ and $|\eta_-|^2$ bilinears are both A_1.

b. A_2 impurity. Next we take an impurity that transforms as A_2, i.e. odd under the vertical mirrors: $V(\boldsymbol{k},\boldsymbol{k}') = iV_0(\boldsymbol{k}\times\boldsymbol{k}')_z\, e^{-|\boldsymbol{k}-\boldsymbol{k}'|^2/q_0^2}$, with $\hat{T} = V\tau_z$ and $q_0 = 20|\boldsymbol{b}_1|$. For s, extended s and $d_{x^2-y^2}$ pairing, $\delta\rho$ vanishes identically at both frequencies. In these states, $\delta\rho$ is a mirror-even observable, so it cannot couple linearly to a mirror-odd perturbation. For $d+id$, $\delta\rho$ also vanishes at $\omega = 0$, but it becomes finite near the gap edge (Fig. 2). The response is purely A_1, so the product $A_2 \otimes A_2$ appears. The second A_2 factor is the chirality $|\eta_+|^2 - |\eta_-|^2$ of the condensate. Reversing the chirality should therefore reverse the sign of $\delta\rho$; we have not yet checked this numerically. An A_2 scatterer thus acts as a direct, sign-sensitive probe of chirality, even without SOC.

B. Rashba spin-orbit coupling with Ansatz superconducting mean fields

We add nearest-neighbour Rashba SOC, $\hat{h}(\boldsymbol{k}) = \varepsilon(\boldsymbol{k}) + \alpha\,(\partial_{k_y}\varepsilon\,\sigma_x - \partial_{k_x}\varepsilon\,\sigma_y)$, with $\alpha = 0.1$ and $\mu = -0.1$. Pairing is a singlet ansatz, $\hat{\Delta} = \Psi(\boldsymbol{k})\, i\sigma_y$, with the same four form factors and $|\Delta| = 0.1$. Figure 3 (left) shows the bands for the chiral state. The impurity is scalar (A_1).

The results for the four states are as follows.

- s, extended s, $d_{x^2-y^2}$: only $\delta\rho_0$ is finite (Fig. 5). All spin channels vanish (below 10^{-11}) at both frequencies. $\delta\rho_0$ is A_1, with an additional E_2 part for $d_{x^2-y^2}$ (weight 15% at $\omega = 0$ and 57% at $\omega = 0.15$; the state is nodal, with four nodes on each helical Fermi surface).

- $d+id$ at $\omega = 0$: also only $\delta\rho_0 \in A_1$.

- $d+id$ at $\omega = 0.2095$ (gap 0.2137): all four channels are finite (Fig. 4, left). $\delta\rho_0$ and $\delta\rho_z$ are real and purely A_1. $\delta\rho_x(\boldsymbol{q})$ and $\delta\rho_y(\boldsymbol{q})$ are purely imaginary and purely E_1. They form the vector pattern $\delta\rho_{x,y}(\boldsymbol{r}) \propto \hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}$, which winds about the impurity. In $\boldsymbol{q}$-space, the maxima are $|\delta\rho_0| = 1.8\times10^3$, $|\delta\rho_x|, |\delta\rho_y| \approx 5\times10^2$ and $|\delta\rho_z| = 2.1\times10^2$, so $\delta\rho_z$ is about 12% of $\delta\rho_0$.

The out-of-plane spin channel $\delta\rho_z$ is the key signal. σ_z is odd under time reversal and under every vertical mirror. An A_1 pattern in $\delta\rho_z$ is therefore allowed only when both

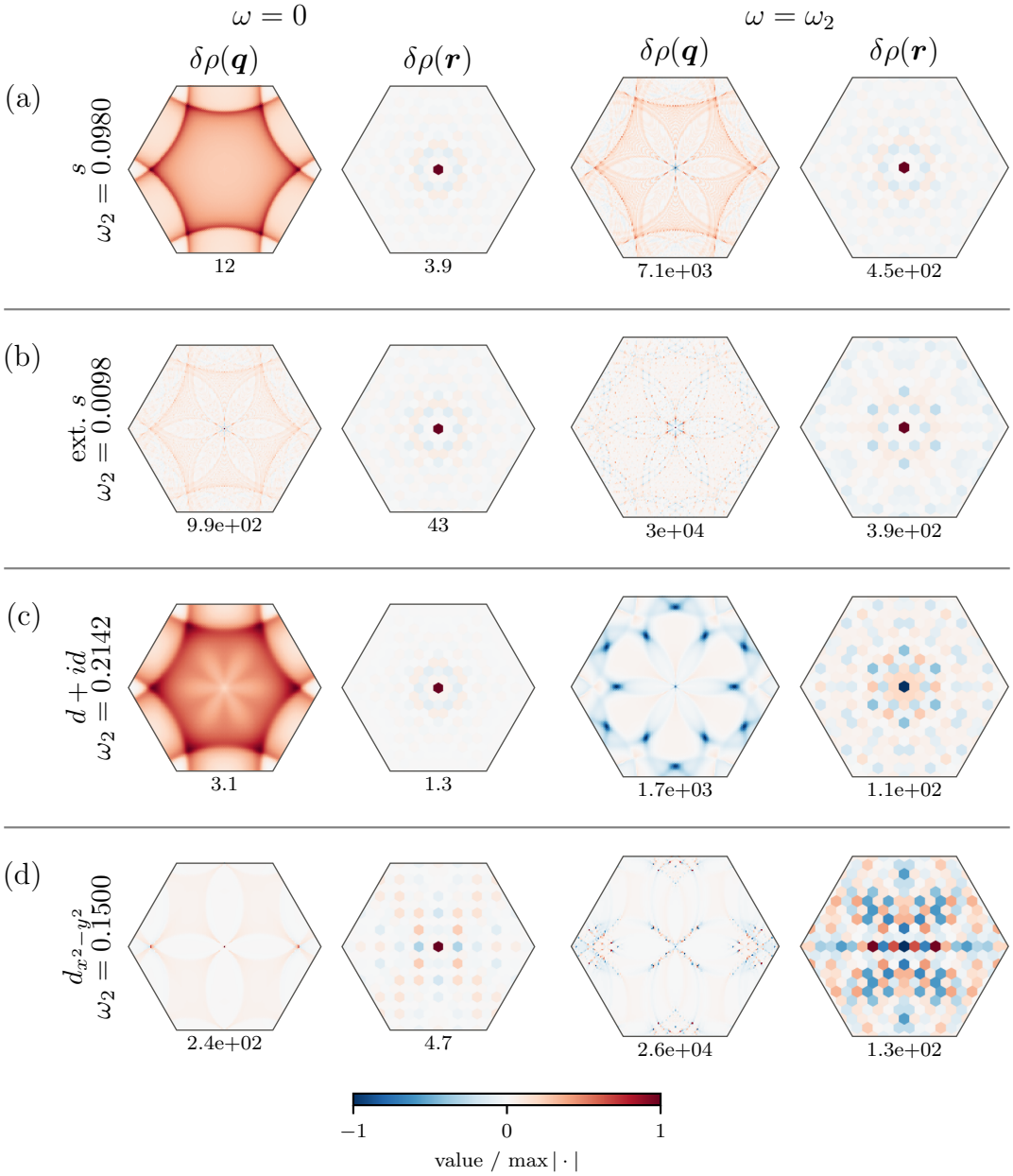


FIG. 1. No SOC, scalar (A_1) impurity, $q_0 = |\boldsymbol{b}_1|$. Rows (a)–(d): s, extended s, $d + id$ and $d_{x^2-y^2}$ pairing. Columns: $\mathrm{Re}\,\delta\rho(\boldsymbol{q})$ and $\delta\rho(\boldsymbol{r})$ at $\omega = 0$, then the same at ω_2, which is given on each row. $\mathrm{Im}\,\delta\rho(\boldsymbol{q})$ and all spin channels vanish. Only the nematic $d_{x^2-y^2}$ state has a component beyond A_1, from the E_2 irrep, which breaks C_6 down to C_2.

symmetries are broken together, which is exactly the chiral superconducting class. It is the spin-resolved analogue of the A_2-impurity response above. Here Rashba SOC locks spin to momentum, so a scalar impurity is enough. The in-plane E_1 texture is less specific: in-plane spin (σ_x, σ_y) already transforms as E_1. In this model it nonetheless appears only in the chiral state.

C. Rashba-Hubbard model with FRG

We take the superconducting mean field from the leading $\boldsymbol{q} = 0$ pairing instability of a divERGe FRG flow for the Rashba-Hubbard model at half filling. The gap is normalised to $\Delta_{\mathrm{rms}} = 0.1$, and the impurity is again scalar.

- $U = 0$, $\alpha = 0.1$: the leading instability is non-degenerate (non-chiral), with gap 0.0777. At $\omega = 0$ and $\omega = 0.0761$, only $\delta\rho_0 \in A_1$ is finite (Fig. 7). All spin channels vanish.

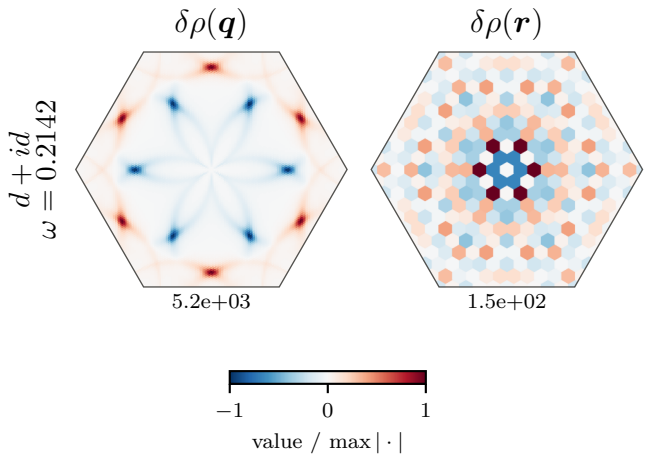


FIG. 2. No SOC, A_2 impurity, $q_0 = 20|\boldsymbol{b}_1|$: the only non-zero case, $d + id$ at $\omega_2 = 0.2142$, with the panel conventions of Fig. 1. $\delta\rho$ vanishes identically for s, extended s and $d_{x^2-y^2}$ pairing at both frequencies, and for $d + id$ at $\omega = 0$. The response is purely A_1.

- $U = 8$, $\alpha = 0.4$: the leading instability is two-fold degenerate. We take the chiral combination $(v_0+iv_1)/\sqrt{2}$, with chirality $\chi = +1$ and gap 0.0863 (Fig. 3, right). The full ansatz pattern appears (Fig. 6, left). $\delta\rho_0, \delta\rho_z \in A_1$ and $\mathrm{Im}\,\delta\rho_{x,y} \in E_1$, each with weight 1. Unlike the ansatz $d + id$ state, the spin signal is already finite at $\omega = 0$, where $\delta\rho_z$ is about 5% of $\delta\rho_0$ (max $|\delta\rho_z(\boldsymbol{q})| = 0.75$ vs. 13.7). At the gap edge ($\omega = 0.0846$), the ratio is also about 5% (1.7×10^3 vs. 3.2×10^4), while $\delta\rho_{x,y}$ reach about 20%.

The chirality signature therefore survives in a microscopically derived gap.

D. Beyond the Born approximation: full T-matrix

For a point-like scalar impurity, the T-matrix can be summed to all orders, $\hat{T}(\omega) = [1 - \hat{V}\hat{G}_{\mathrm{loc}}(\omega)]^{-1}\hat{V}$, with $\hat{V} = V_0\tau_z$ and $\hat{G}_{\mathrm{loc}}(\omega) = N^{-1}\sum_{\boldsymbol{k}} \hat{g}_0(\boldsymbol{k}, \omega)$. It is independent of $\boldsymbol{q}$, so $S(\boldsymbol{q})$, $\delta\rho_\mu$ and the irrep analysis are otherwise unchanged. A point impurity needs no momentum cutoff, so none is applied in this section ($q_0 \to \infty$). Born values quoted here for comparison are for the same point impurity; the cutoff is fully symmetric (A_1), so it changes amplitudes but never the irreps. Besides $V_0 = 0.1$, we also use resonant strengths $V_0 = 1/\lambda$, where λ is a real eigenvalue of $\hat{G}_{\mathrm{loc}}(\omega, \Gamma = 0)\,\tau_z$. At such a V_0, the impurity has a bound state at ω, and $\hat{T}$ is limited only by Γ. If all eigenvalues λ are complex, there is no bound state at ω. This is the case for s pairing (Anderson's theorem) and for the non-chiral FRG state. A bound state needs an energy gap, so resonances are only considered for gapped states, at ω inside the gap. They are not considered for the nodal $d_{x^2-y^2}$ state. Every resonant case is listed in Table II. All other parameters are as above. At $V_0 = 0.1$, the full $\hat{T}$ leaves every entry of Table I unchanged, with or without SOC, except for the Rashba $d + id$ state at $\omega = 0$ discussed below.

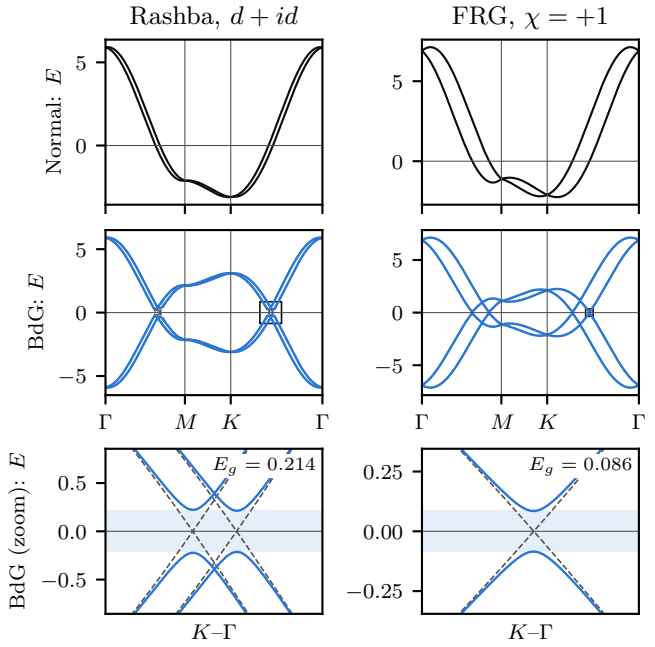


FIG. 3. Bands on Γ–M–K–Γ for the two chiral states. Left: Rashba model ($\alpha = 0.1$, $\mu = -0.1$) with $d+id$ singlet pairing, $|\Delta| = 0.1$. Right: FRG gap of the Rashba-Hubbard model ($U = 8$, $\alpha = 0.4$, half filling), $\chi = +1$, $\Delta_{\mathrm{rms}} = 0.1$. Rows: normal state; BdG bands (solid) with $\pm\varepsilon(\boldsymbol{k})$ (dashed); BdG bands zoomed on the boxed region, the Fermi crossing with the smallest gap on the path. The shading marks $|E| < E_g$, with E_g the gap over the whole Brillouin zone.

a. Ansatz pairings (Rashba).

- s, extended s, $d_{x^2-y^2}$: the spin channels still vanish for every $\hat{T}$, including at resonance. $\delta\rho_0$ keeps the irreps it has in the Born approximation. For $d_{x^2-y^2}$ this is A_1, with an E_2 weight of 17% at $\omega = 0$ and 59% at $\omega = 0.15$ for $V_0 = 0.1$. The extended-s gap-edge resonances are weak, with max $|\hat{T}| = 9.3$ and 5.8 (Table II). The $d_{x^2-y^2}$ state has no gap, so it has no resonances.

- $d+id$ at $\omega = 0$: in the Born approximation, only $\delta\rho_0$ is finite. With the full $\hat{T}$, the spin pattern of the gap edge appears here too: $\delta\rho_z \in A_1$ and $\mathrm{Im}\,\delta\rho_{x,y} \in E_1$, each with weight 1. It is weak at $V_0 = 0.1$, with max $|\delta\rho_z(\boldsymbol{q})| = 2.3 \times 10^{-3}$ and $|\delta\rho_{x,y}| = 4.1 \times 10^{-3}$, against 3.7 for $\delta\rho_0$. At the zero-energy resonance, $V_0 = 6.28$, it grows to $\delta\rho_z/\delta\rho_0 \approx 0.4\%$. There, $\delta\rho_0$ is dominated by the localised bound state and is almost featureless in $\boldsymbol{q}$, while the spin channels keep their $V_0 = 0.1$ patterns.

- $d + id$ at $\omega = 0.2095$ (Fig. 4): the irreps are un-

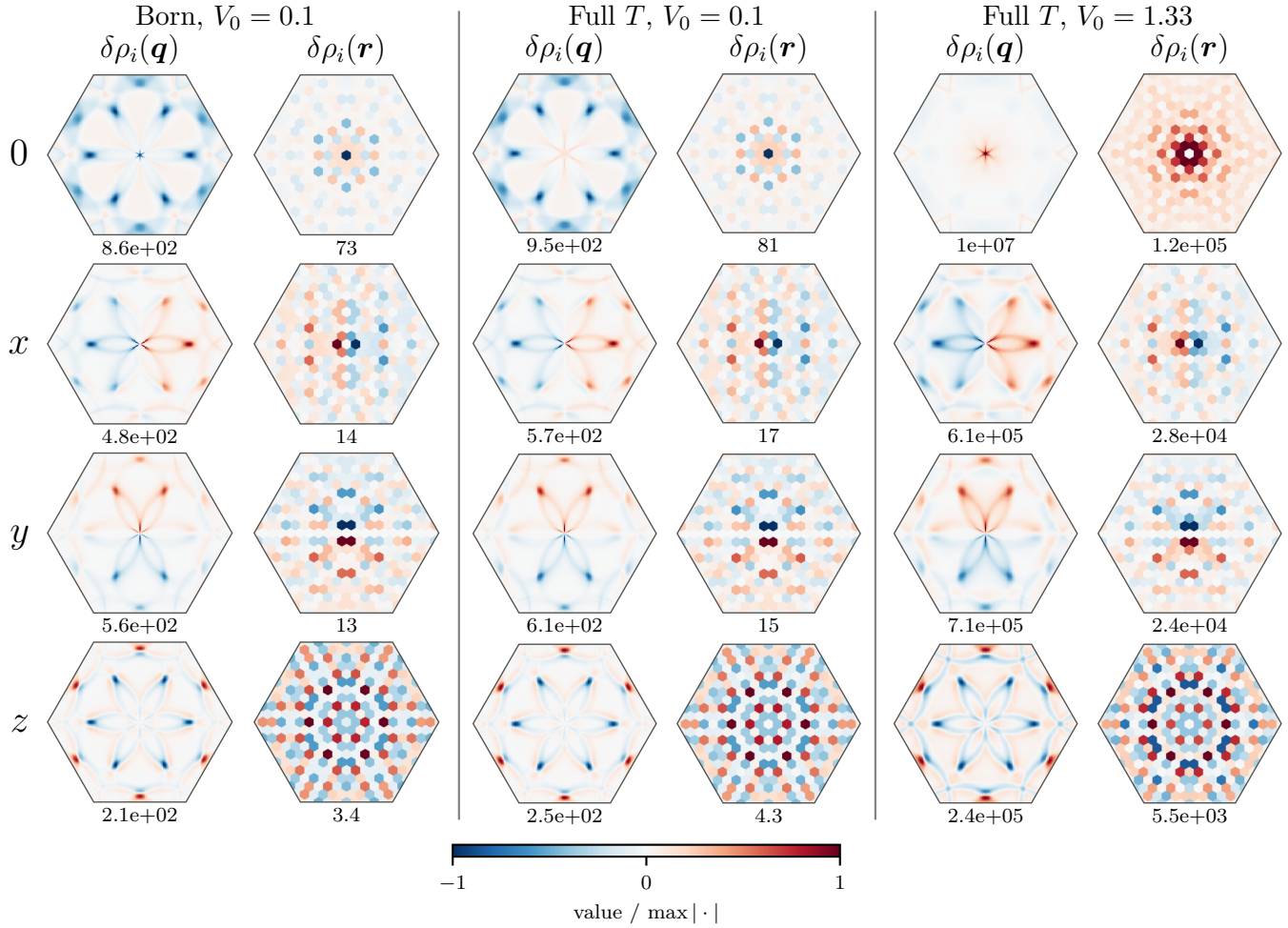


FIG. 4. Spin-resolved QPI for the Rashba model with $d + id$ pairing and a scalar impurity, at $\omega_2 = 0.2095$. Left: Born approximation, $q_0 = |\boldsymbol{b}_1|$. Centre: full T-matrix at the same V_0. Right: full T-matrix at the resonance $V_0 = 1.33$ ($\max|\hat{T}| = 40$). Here $\delta\rho_0(\boldsymbol{q})$ is dominated by a broad peak at $\boldsymbol{q} = 0$ from the impurity bound state, which hides the scattering pattern of the other two columns; the spin channels keep that pattern. The full-T panels have no q cutoff. Rows are $i = 0, x, y, z$; the $\boldsymbol{q}$-maps show $\mathrm{Im}\,\delta\rho_i$ for x, y and $\mathrm{Re}\,\delta\rho_i$ for $0, z$, and the other part vanishes. Each panel is normalised to its own maximum, given below it; for the $\boldsymbol{q}$-maps this is the maximum over $\boldsymbol{q} \neq 0$, so the $\boldsymbol{q} = 0$ point saturates. The irreps are given in Table I.

changed. At $V_0 = 0.1$, $\delta\rho_z$ grows to 27% of $\delta\rho_0$ (254 vs. 951), from 13% in the Born approximation. At the two resonances, $V_0 = -2.28$ and 1.33, $\delta\rho_0$ dominates and the ratio drops to 10^{-4} and 1% respectively. At $V_0 = 1.33$, $\hat{T} = t_0 + t_z\tau_z$ with $|t_0| \approx |t_z| \approx 20$; the τ_0 part gives the bound-state peak in $\delta\rho_0$, about 20 times the τ_z part, while the τ_z part alone reproduces the Born pattern of $\delta\rho_0$ (overlap 0.91).

b. FRG pairings.

- $U = 0$, $\alpha = 0.1$: there are no bound states, and only $\delta\rho_0 \in A_1$ is finite, as in the Born approximation (Fig. 7, right).
- $U = 8$, $\alpha = 0.4$, chiral ($\chi = +1$): the same irreps as in the Born approximation, each with weight 1. At $\omega = 0$ and $V_0 = 0.1$, $\delta\rho_z/\delta\rho_0 = 5.0\%$, unchanged from Born. At the gap edge, the ratio rises from 5.8% to 15% (2.0×10^3 vs. 1.3×10^4), and $\delta\rho_{x,y}$ reach about 55% of $\delta\rho_0$ (Fig. 6, centre). There are two resonances at $\omega = 0$ ($V_0 = 5.38, 6.56$) and four at the gap edge, compared with one and two for the ansatz $d+id$. At these resonances $\delta\rho_z/\delta\rho_0 = 4$–$7\%$ (Table II).

Going beyond Born therefore never adds an irrep to a channel; it only changes the relative amplitudes. The chirality signature $\delta\rho_z \in A_1$ survives at all orders. For the ansatz $d + id$ state, it also extends to $\omega = 0$.

E. Sn/Si(111) Wannier model

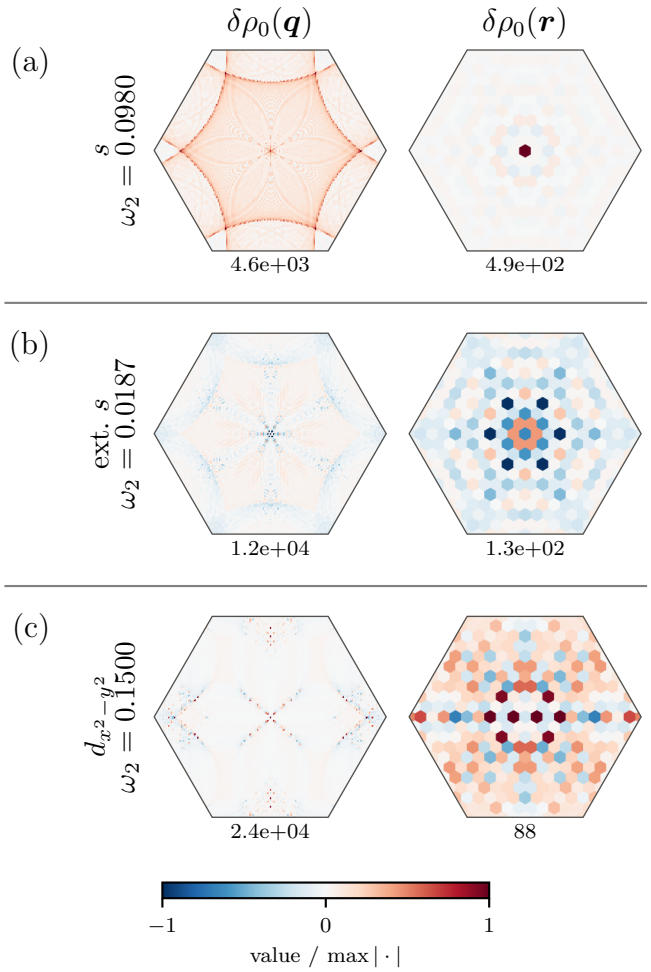


FIG. 5. $\mathrm{Re}\,\delta\rho_0$ for the Rashba model with s, extended s and $d_{x^2-y^2}$ pairing, scalar impurity, Born approximation, $q_0 = |\boldsymbol{b}_1|$, each at its ω_2. All spin channels vanish. Normalisation as in Fig. 4.

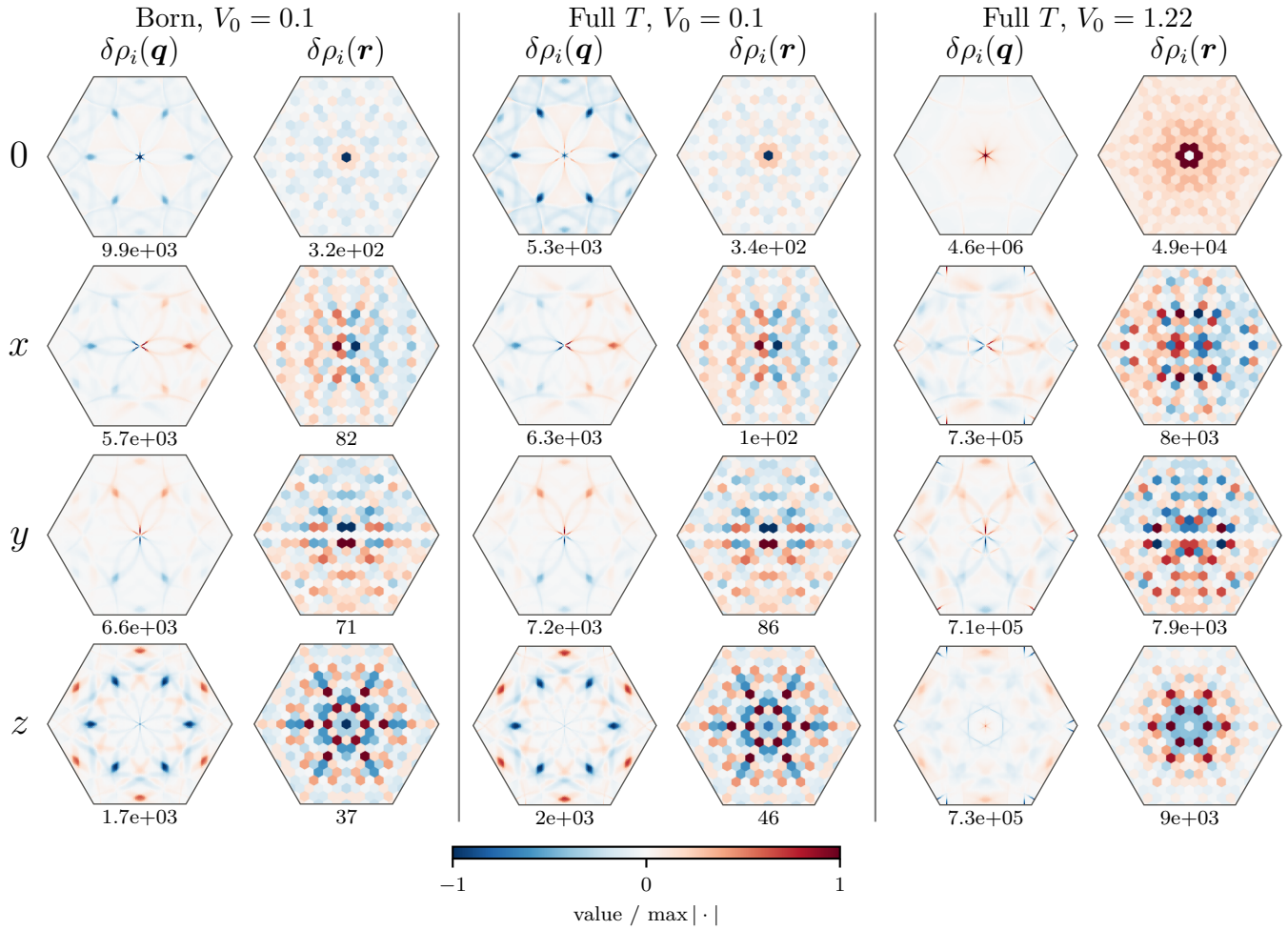


FIG. 6. Spin-resolved QPI for the chiral FRG state of Fig. 3, scalar impurity, $\omega = 0.0846$. Left: Born approximation, $q_0 = |\boldsymbol{b}_1|$. Centre: full T-matrix at the same V_0. Right: full T-matrix at the resonance $V_0 = 1.22$ ($\max|\hat{T}| = 12$), the sharpest of the four at this ω; as in Fig. 4, the bound state dominates $\delta\rho_0$, while the spin channels keep their pattern. The full-T panels have no q cutoff. Panel conventions as in Fig. 4; $\delta\rho_z$ is A_1 in all three.

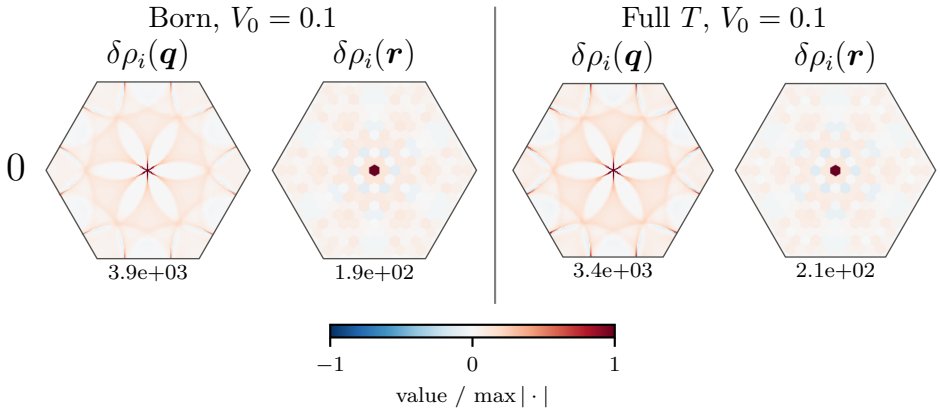


FIG. 7. $\mathrm{Re}\,\delta\rho_0$ for the non-chiral FRG state ($U = 0$, $\alpha = 0.1$), scalar impurity, $\omega = 0.0761$, as in Fig. 6. All spin channels vanish. This state has no impurity bound states, so there is no resonant case.

TABLE II. Full T-matrix at the resonant impurity strengths $V_0 = 1/\lambda$, scalar impurity. $\max|\hat{T}|$, at $\Gamma = 10^{-3}$, measures how sharp the resonance is. Ratios are of the $\boldsymbol{q}$-space maxima, including $\boldsymbol{q} = 0$; 0 means numerically zero. In every row $\delta\rho_0, \delta\rho_z \in A_1$ and $\mathrm{Im}\,\delta\rho_{x,y} \in E_1$, each with weight 1, where finite. Only states with SOC were searched. s pairing and the $U = 0$ FRG state have no resonances; the nodal $d_{x^2-y^2}$ state has no gap, so resonances are not considered.

State	ω	V_0	$\max\lvert\hat{T}\rvert$	$\delta\rho_z/\delta\rho_0$	$\delta\rho_{x,y}/\delta\rho_0$
$d + id$ (Rashba)	0.0000	6.28	783	3.7×10^{-3}	6.7×10^{-3}
	0.2095	−2.28	36.8	1.0×10^{-4}	2.4×10^{-4}
	0.2095	1.33	40	1.3×10^{-2}	3.8×10^{-2}
ext. s (Rashba)	0.0187	−2.96	5.76	0	0
	0.0187	1.60	9.29	0	0
FRG $U = 8$, $\chi = +1$	0.0000	5.38	262	4.0×10^{-2}	5.6×10^{-2}
	0.0000	6.56	262	4.1×10^{-2}	5.2×10^{-2}
	0.0846	−1.96	9.54	5.3×10^{-2}	5.4×10^{-2}
	0.0846	−1.72	8.86	3.7×10^{-2}	1.5×10^{-2}
	0.0846	1.12	10.9	3.7×10^{-2}	6.2×10^{-2}
	0.0846	1.22	11.9	7.0×10^{-2}	6.9×10^{-2}